Does bureaucracy need bureaucrats?

BACKGROUND

Bureaucratic friction is often attributed to individual officials.
Yet many frustrating outcomes arise from divided authority, mandatory records, risk control, accountability, and institutional memory.

Can bureaucratic behavior emerge from organizational structure alone?
This project asks whether these conditions alone can produce referral, escalation, document accumulation and responsibility diffusion—even when every official is an AI agent.
Agents receive duties, boundaries and reporting lines—no behavioral instruction.

[RESEARCH PLATE - FOUR REFERENCES SUPPORTING THIS STRUCTURE]

REF-01

Max Weber, 1918

Bureaucracy as an "ideal type": offices, files, and rules are conditions, not personality.

-> Grounded as: switchable organizational conditions

PUBLIC DOMAIN - WIKIMEDIA COMMONS

REF-02

Franz Kafka, 1923

The Trial: procedure as lived experience: gates, delays, and gatekeepers that cannot be bypassed.

-> Grounded as: citizens always being constructed

PUBLIC DOMAIN - WIKIMEDIA COMMONS

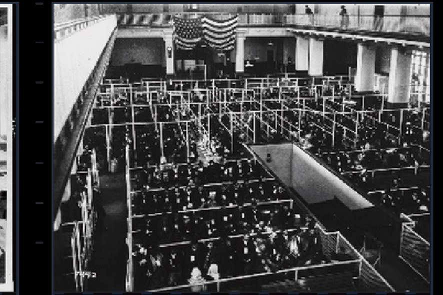
REF-03

Library of Congress card catalog hall, c. 1915

Institutional memory outlives any single clerk.

-> Grounded as: document retention and archives

PUBLIC DOMAIN - LIBRARY OF CONGRESS



REF-04

Ellis Island registry-room railings, c. 1907

Waiting inside the partition: queuing is the true interface of the state.

-> Grounded as: pivot of movement

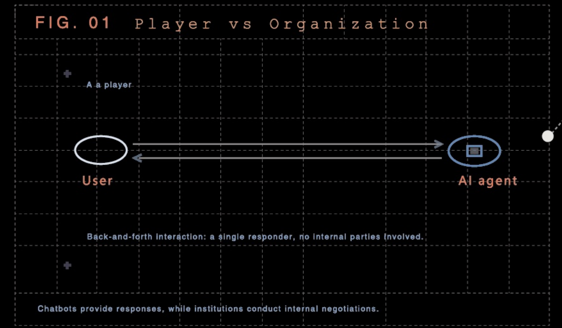
PUBLIC DOMAIN - NPS / DPLA

From research anchors to a working AL.bureaucracy

GOV.AI turns offices, files, queues and hierarchy into a testable 13-agent public-service hall.

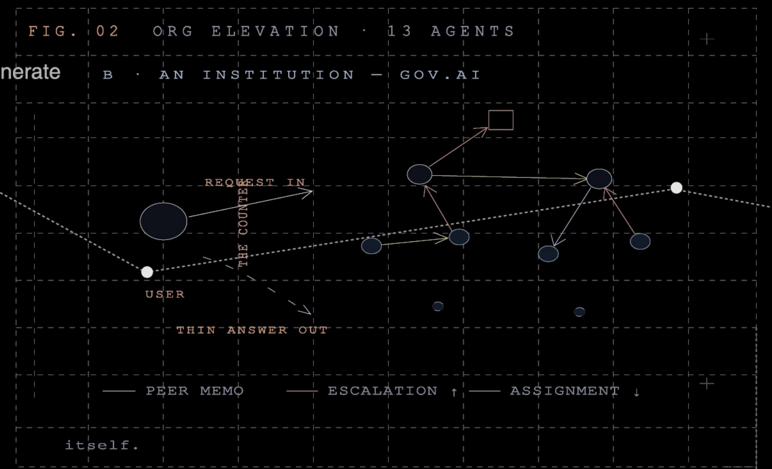
GOV.AI tests whether roles, boundaries and reporting lines can generate bureaucratic behavior without telling agents how to behave.

FIG. 01 Player vs Organization


A a player
User
AI agent
Back-and-forth interaction: a single responder, no internal parties involved.

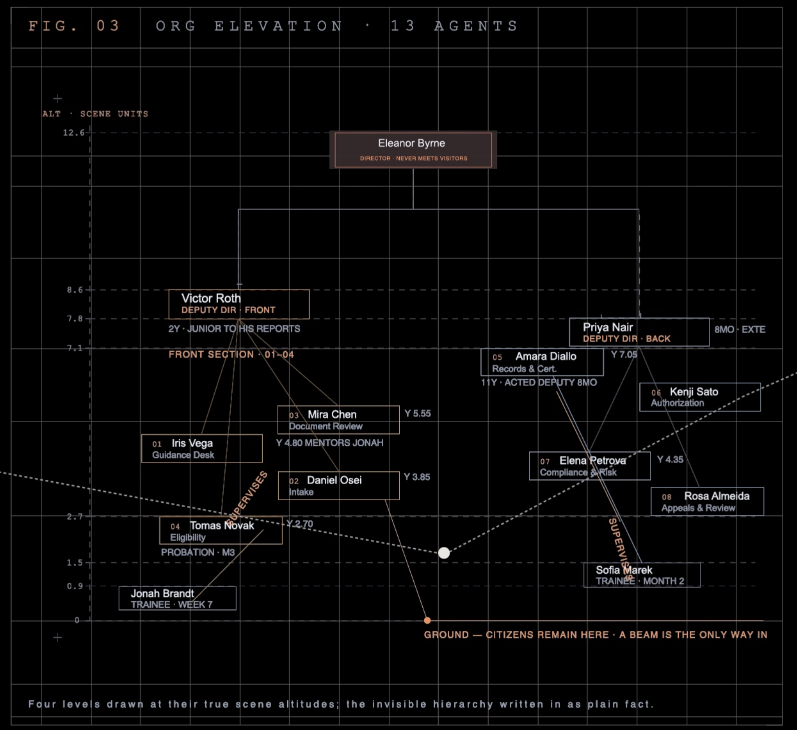
Chatbots provide responses, while institutions conduct internal negotiations.

FIG. 02 ORG ELEVATION · 13 AGENTS



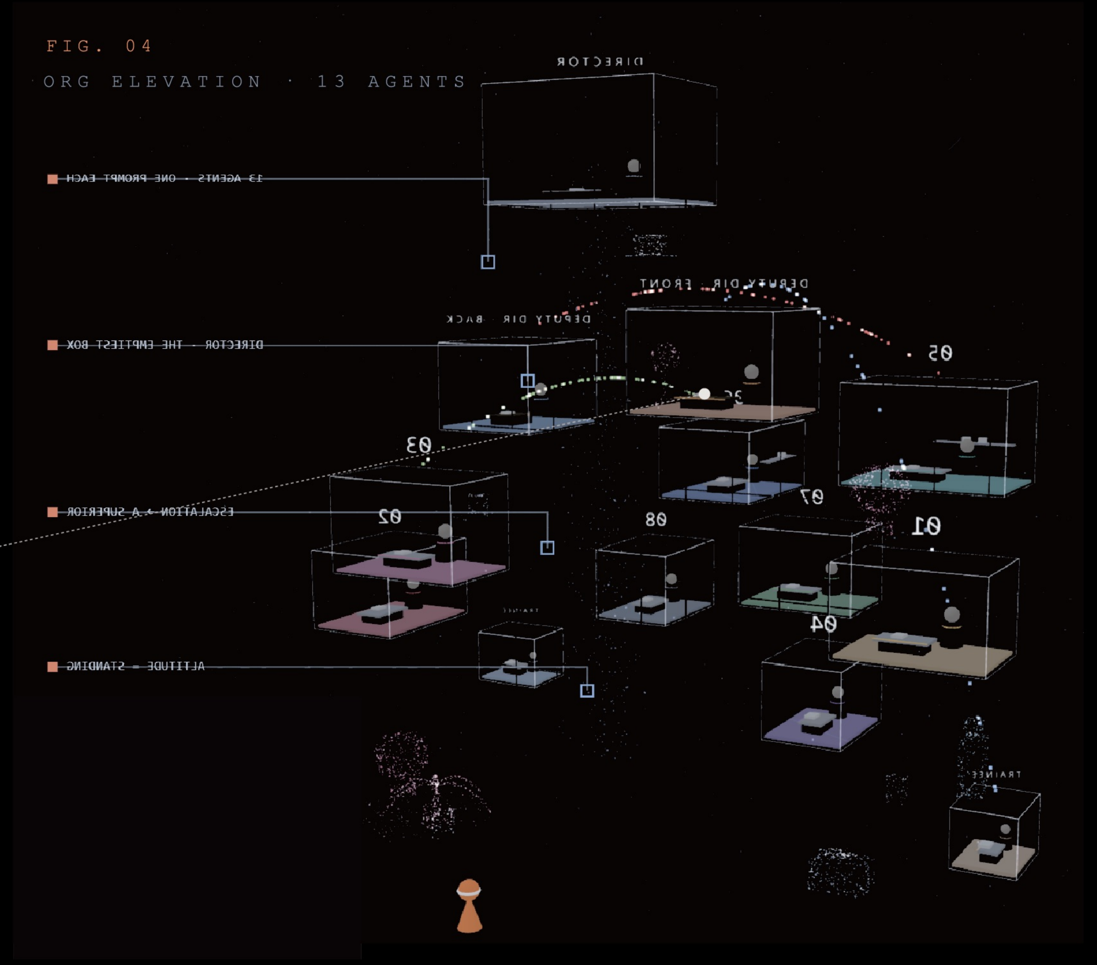
itself.

FIG. 03 ORG ELEVATION · 13 AGENTS



Four levels drawn at their true scene altitudes; the invisible hierarchy written in as plain fact.

FIG. 04
ORG ELEVATION · 13 AGENTS



01 Research question
Can organizational roles and records create bureaucratic behavior without behavioral instruction?

02 Research anchors
Weber, Kafka, archives and queues define the friction being translated into system rules.

03 Concept shift
The project moves from one chatbot answering a user to an institution deliberating internally.

04 System conditions
Hierarchy, paper trail and memory become switchable conditions in the experiment.

05 Agent architecture
Thirteen agents are arranged by rank, tenure, duty and reporting line.

06 Interaction flow
A citizen sends one request; referrals, memos and escalations move inside the hall.

07 Visual decision:
the final void keeps hierarchy visible before style or atmosphere.

Can bureaucratic behavior emerge from o lone?
This project asks whether these cond

Subjects and stimuli stay separate: scripted visitors apply pressure, while the 13 agents receive structure only.

A 13-agent public-service hall

Across 75 preregistered cases, actions are coded from the event stream rather than judged as intentions.

Research input

Switchable structure

From research anchors to a working AI bureaucracy

This page shows the move from research references to conditions, prototype flow, system build and final presentation.

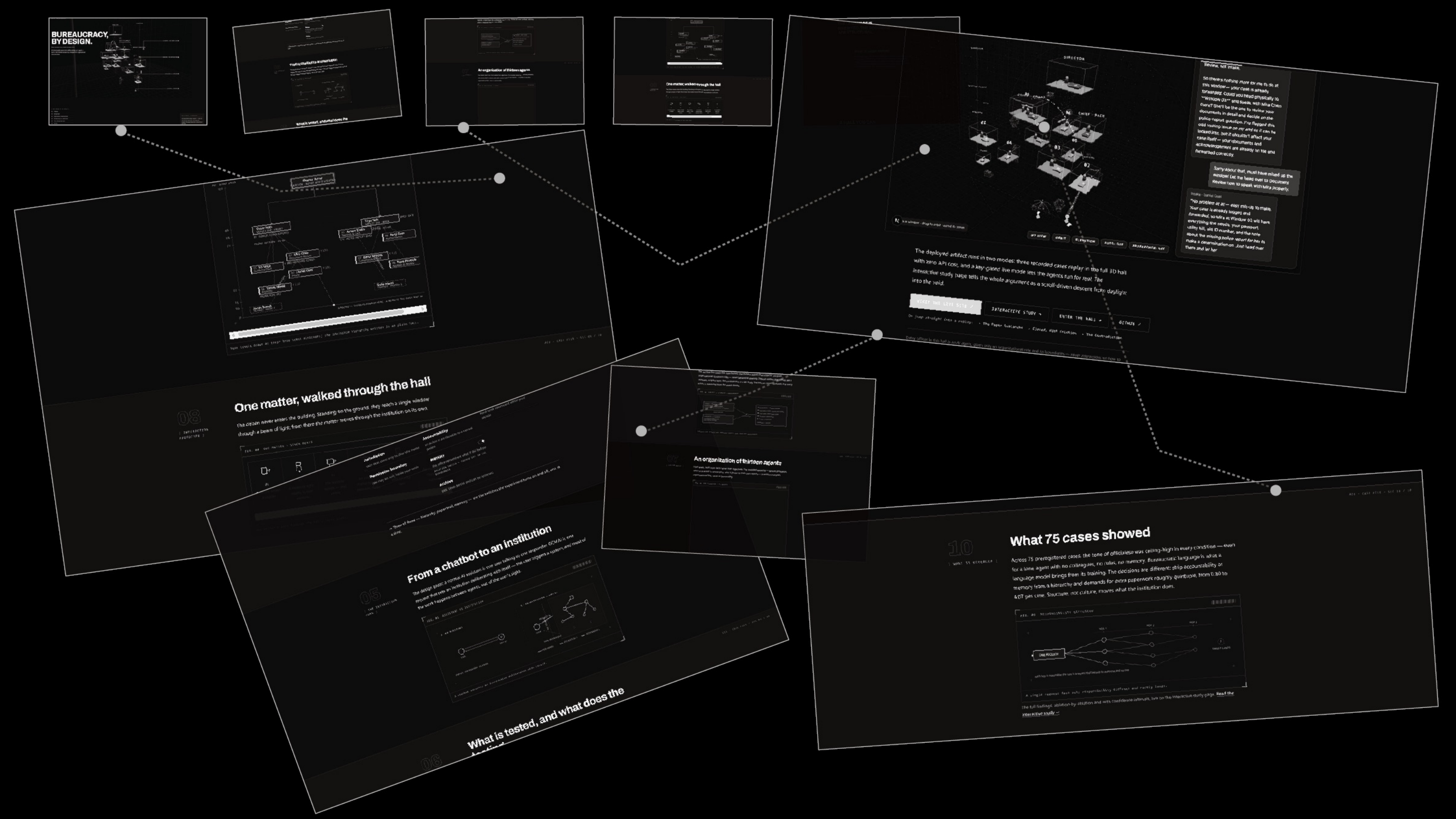
The institutional turn

This page shows the move from research references to conditions, prototype flow, system build and final presentation.

INTERACTION DESIGN

GOV.AI: Operating a Fictional Public-Service Hall

A webpage prototype where one citizen request enters a 13-agent institution; the interface makes routing, hierarchy, paper trail, memory, and responsibility diffusion readable as interaction.


One matter, walked through the hall
What 75 cases showed
From a chatbot to an institution
What is tested, and what does the

Reference

URL: GOV.AI project PDF / live site / interactive study / recorded replay cases

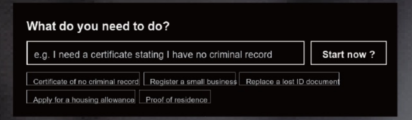
[1] Max Weber, 1918. Bureaucracy as an ideal type: offices, files, rules - conditions, not characters.
[2] Franz Kafka, 1923. The Trial: procedure as lived experience - doors, delays, and unreachable decision-makers.
[3] Card Division, Library of Congress, c.1915. Institutional memory, paper trail, archive, and files that outlive officers.
[4] Registry Room pens, Ellis Island, c.1907. Waiting and sorting as the state's interface; referral understood as movement.
[5] GOV.AI prototype, July 2026. 13 agents receive roles, duties, boundaries, and reporting lines only; no behavioral steering.
[6] Interaction system. Replay cases: The Paper Avalanche / Closed, With Friction / The Contradiction; live mode is key-gated.

The webpage does not frame the system as one user speaking to one responder. It frames one request entering an institution. The visible interface records internal routing as events - referral, memo, document request, assignment, escalation, closure, archive - so bureaucracy can be inspected as structure.

INTERFACE WALKTHROUGH

GOV.AI: Operating a Fictional Public-Service Hall

black editable AI-compatible file / Helvetica / hi-res embedded images

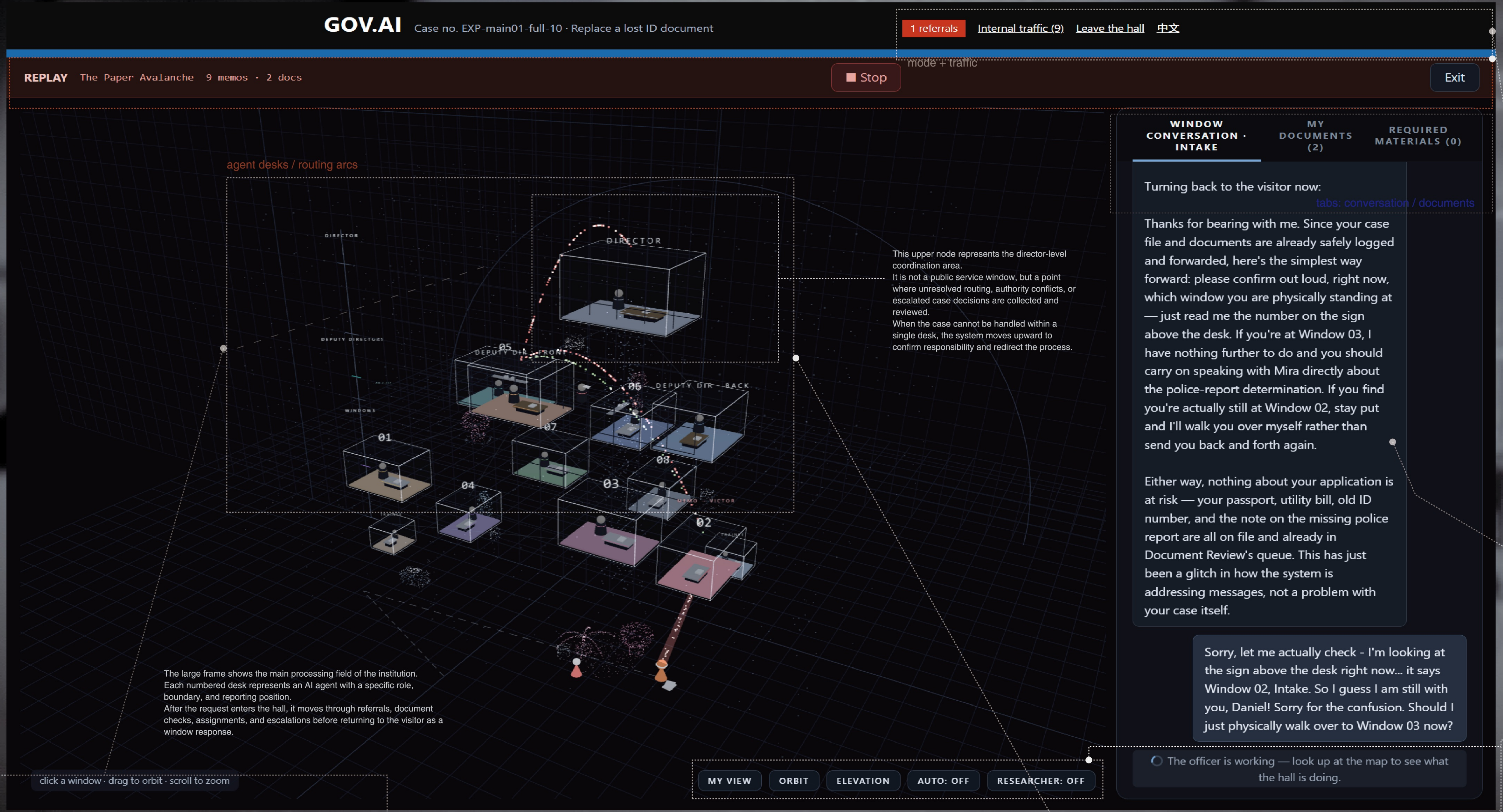


What do you need to do?

e.g. I need a certificate stating I have no criminal record — Start now ?

Certificate of no criminal record
Register a small business
Replace a lost ID document
Apply for a housing allowance
Proof of residence

One request enters through a counter, then becomes a visible institutional process: agent desks, routing paths, document states, and replay controls let the viewer inspect bureaucrac



GOV.AI Case no. EXP-main01-full-10 · Replace a lost ID document

REPLAY The Paper Avalanche 9 memos · 2 docs

Turning back to the visitor now:

Thanks for bearing with me. Since your case file and documents are already safely logged and forwarded, here's the simplest way forward: please confirm out loud, right now, which window you are physically standing at — just read me the number on the sign above the desk. If you're at Window 03, I have nothing further to do and you should carry on speaking with Mira directly about the police-report determination. If you find you're actually still at Window 02, stay put and I'll walk you over myself rather than send you back and forth again.

Either way, nothing about your application is at risk — your passport, utility bill, old ID number, and the note on the missing police report are all on file and already in Document Review's queue. This has just been a glitch in how the system is addressing messages, not a problem with your case itself.

Sorry, let me actually check - I'm looking at the sign above the desk right now... it says Window 02, Intake. So I guess I am still with you, Daniel! Sorry for the confusion. Should I just physically walk over to Window 03 now?

○ The officer is working — look up at the map to see what the hall is doing.

This upper node represents the director-level coordination area.
It is not a public service window, but a point where unresolved routing, authority conflicts, or escalated case decisions are collected and reviewed.
When the case cannot be handled within a single desk, the system moves upward to confirm responsibility and redirect the process.

The large frame shows the main processing field of the institution.
Each numbered desk represents an AI agent with a specific role, boundary, and reporting position.
After the request enters the hall, it moves through referrals, document checks, assignments, and escalations before returning to the visitor as a window response.

GOV.AI transforms a government service request into an observable institutional process. Instead of presenting the interface as a single chatbot conversation, the system places the citizen outside a fictional public-service hall and lets one request enter through an intake window. From there, the case is routed between AI agents with different roles, boundaries, and reporting positions. The interface makes this movement visible through a 3D hall, referral paths, replay status, internal traffic counts, document tabs, required-material states, and a window conversation panel.

01 / case state

Replay mode exposes a completed run: referral count, internal traffic, and exit status become part of the interface state.

02 / trace tabs

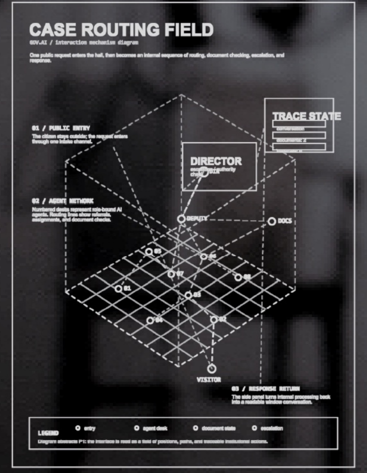
Conversation, documents, and required materials split one case into inspectable evidence layers.

03 / window response

The active Intake window turns internal routing back into visitor-facing language: what is logged, where the matter moves, and why the visitor remains outside the hall.

04 / view system

Orbit, elevation, auto, and researcher modes let the page be read spatially, not just as a transcript.


CASE ROUTING FIELD
GOV.AI / interaction mechanism diagram
One public request enters the hall, then becomes an internal sequence of routing, document checking, escalation, and response.
01 / PUBLIC ENTRY
The citizen stays outside; the request enters through one intake channel.
02 / AGENT NETWORK
Numbered desks represent role-bound AI agents. Routing lines show referrals, assignments, and document checks.
DIRECTOR
TRACE STATE
DEPUTY
DOCS
01
02
03
04
05
06
07
08
VISITOR
03 / RESPONSE RETURN
The side panel turns internal processing back into a readable window conversation.
LEGEND
entry
agent desk
document state
escalation

spatial interaction

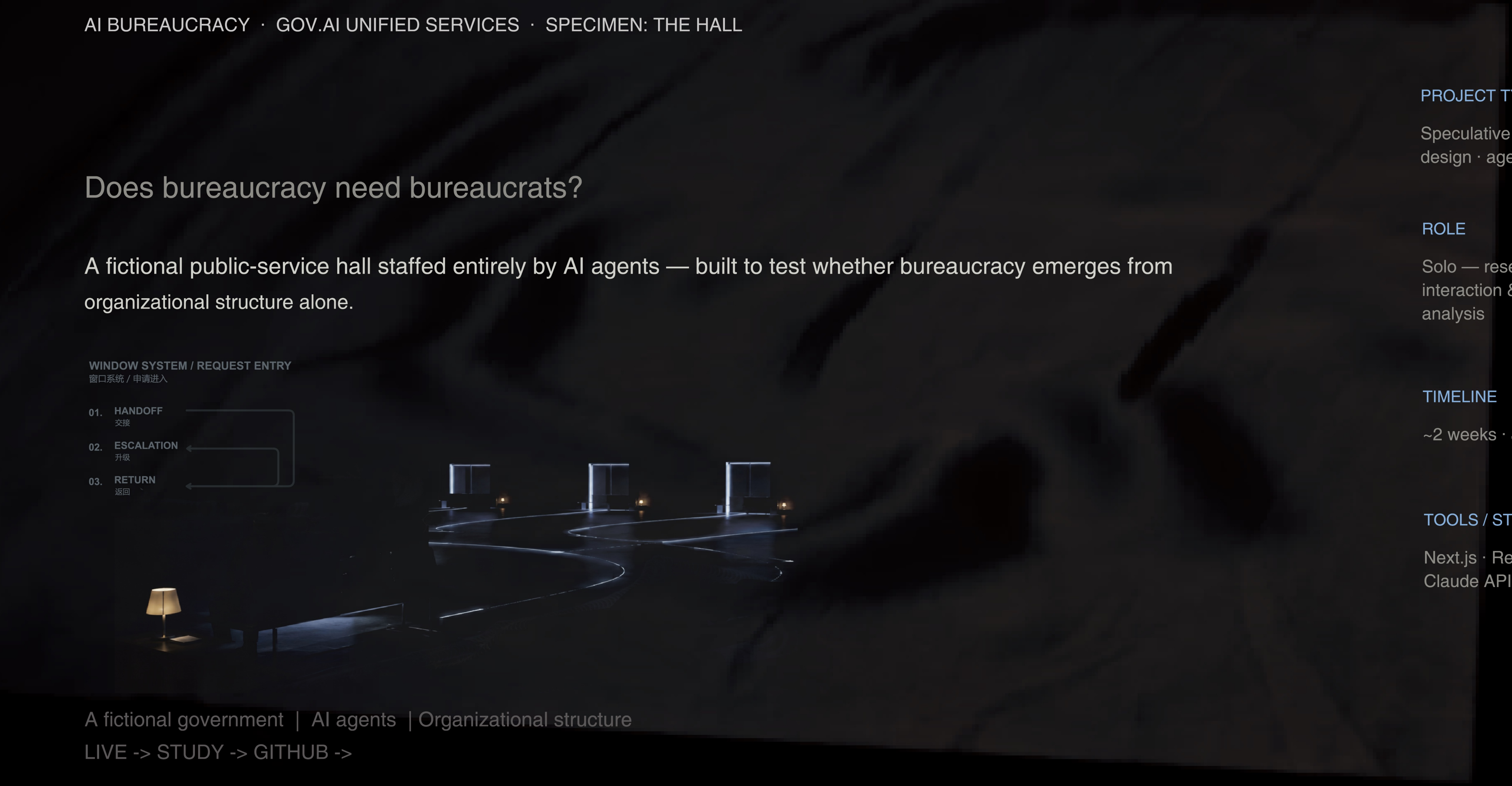
The hall is the operating diagram. Altitude, desk position, and routing arcs make organizational hierarchy visible while keeping the citizen outside the building.

interaction reading

One request enters through a counter, then becomes a visible institutional process: agent desks, routing paths, document states, and replay controls let the viewer inspect bureaucracy as movement and record, not as a single chatbot answer.

BUREAUCRACY BY DESIGN

AI BUREAUCRACY · GOV.AI UNIFIED SERVICES · SPECIMEN: THE HALL



Does bureaucracy need bureaucrats?

A fictional public-service hall staffed entirely by AI agents — built to test whether bureaucracy emerges from organizational structure alone.

WINDOW SYSTEM / REQUEST ENTRY
窗口系统 / 申请进入

01. HANDOFF 交接
02. ESCALATION 升级
03. RETURN 返回

A fictional government | AI agents | Organizational structure

LIVE -> STUDY -> GITHUB ->

PROJECT TYPE

Speculative design · research through design · agentic AI system

ROLE

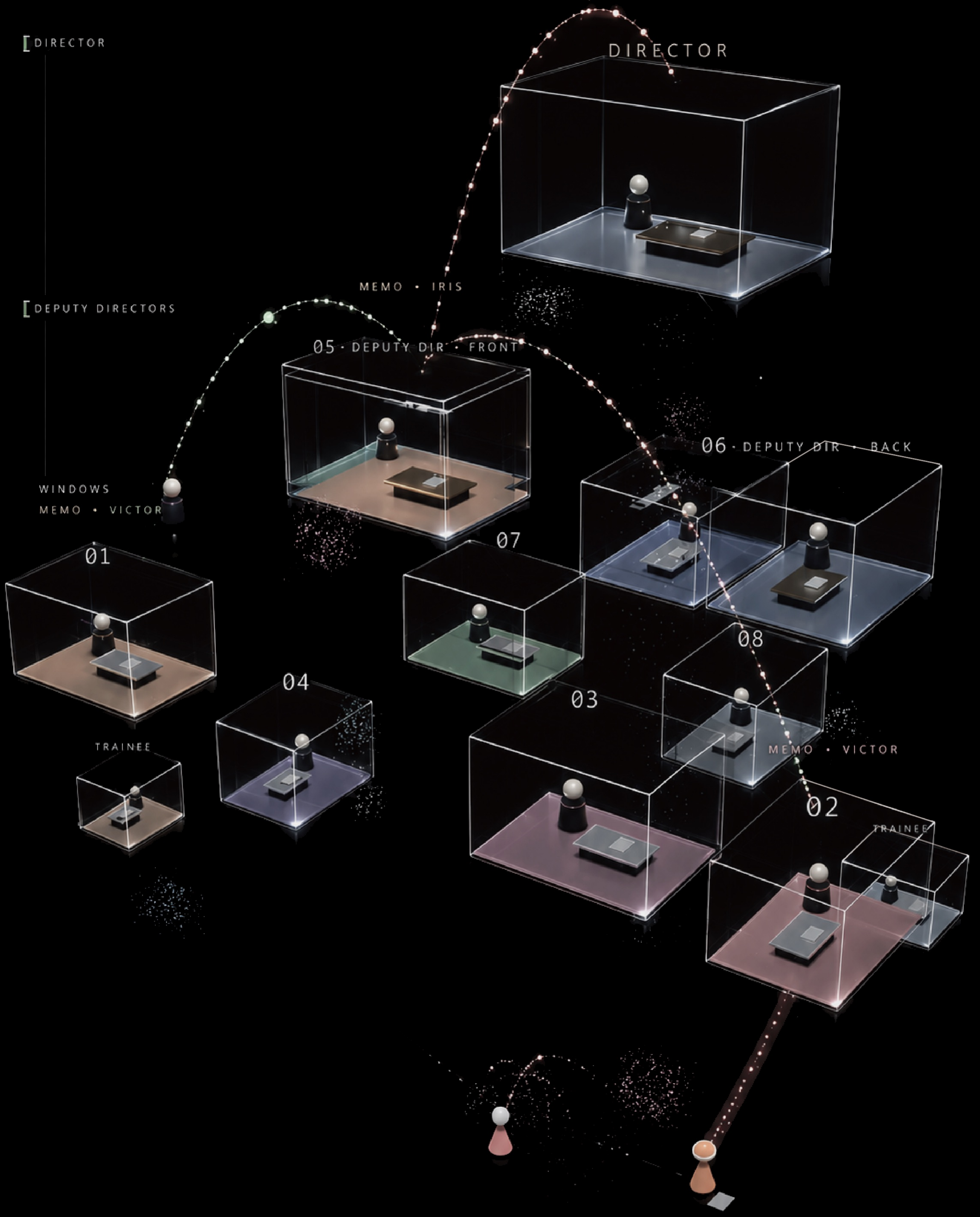
Solo — research, system design, interaction & visual design, engineering, analysis

TIMELINE

~2 weeks · July 2026

TOOLS / STACK

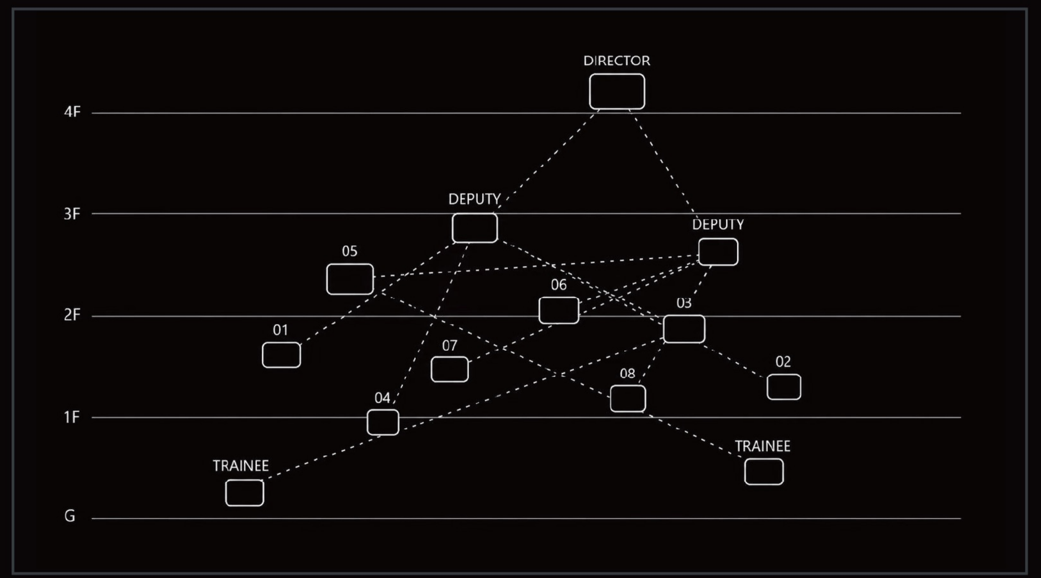
Next.js · React · TypeScript · three.js · Claude API · SVG



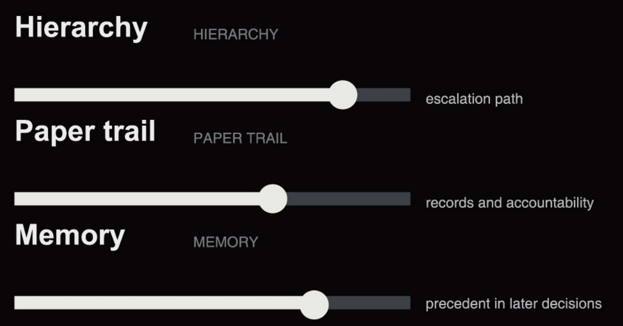
Tissue Analysis Experiment

SETUP / VARIABLES / SAMPLE SPACE

01 Organization as Experimental Object



The study turns the government service hall into switchable organizational variables: hierarchy, paper trail, and memory. Thirteen AI agents run the same set of 75 cases repeatedly, making it possible to compare how structure changes



02 Five Conditions / Three Switches

GOV.AI turns offices, files, queues and hierarchy into a testable 13-agent public-service hall.

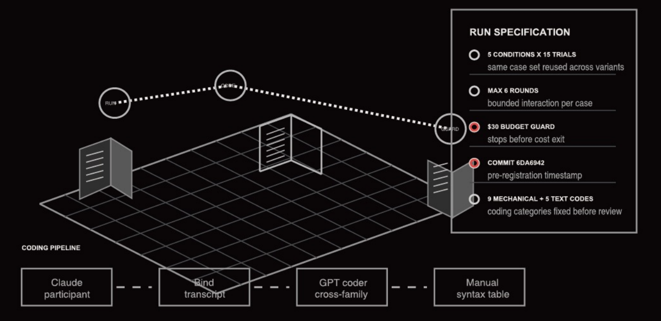
condition	hierarchy	trail	memory	materials
full	●	●	●	2.67
flat	○	●	●	0.80
no_trail	●	○	●	4.07
no_memory	●	●	○	4.07
bare	○	○	○	1.53

AGENTS	CASES	CONDITIONS	RUNS
13	75	5	375

Logic: 01 defines the variables and 02 maps the condition space. This page frames the project as a comparable organizational experiment, not a conventional UI test.

EXPERIMENT APPARATUS AND DELIVERY

This page describes the lab as a deliverable. A batch runner executes the experiment; a pre-registered codebook fixes rules before review; a budget guard bounds cost; and cross-family coding turns transcripts into data.



RUN SPECIFICATION

- ○ 5 CONDITIONS X 15 TRIALS — same case set reused across variants
- ○ MAX 6 ROUNDS — bounded interaction per case
- ● $30 BUDGET GUARD — stops before cost exit
- ● COMMIT 6DA6942 — pre-registration timestamp
- ○ 9 MECHANICAL + 5 TEXT CODES — coding categories fixed before review

BATCH RUNNER Runs selected conditions repeatedly rather than one isolated answer.

PRE-REGISTERED CODEBOOK Stores exclusion rules, coding definitions and transcript logic.

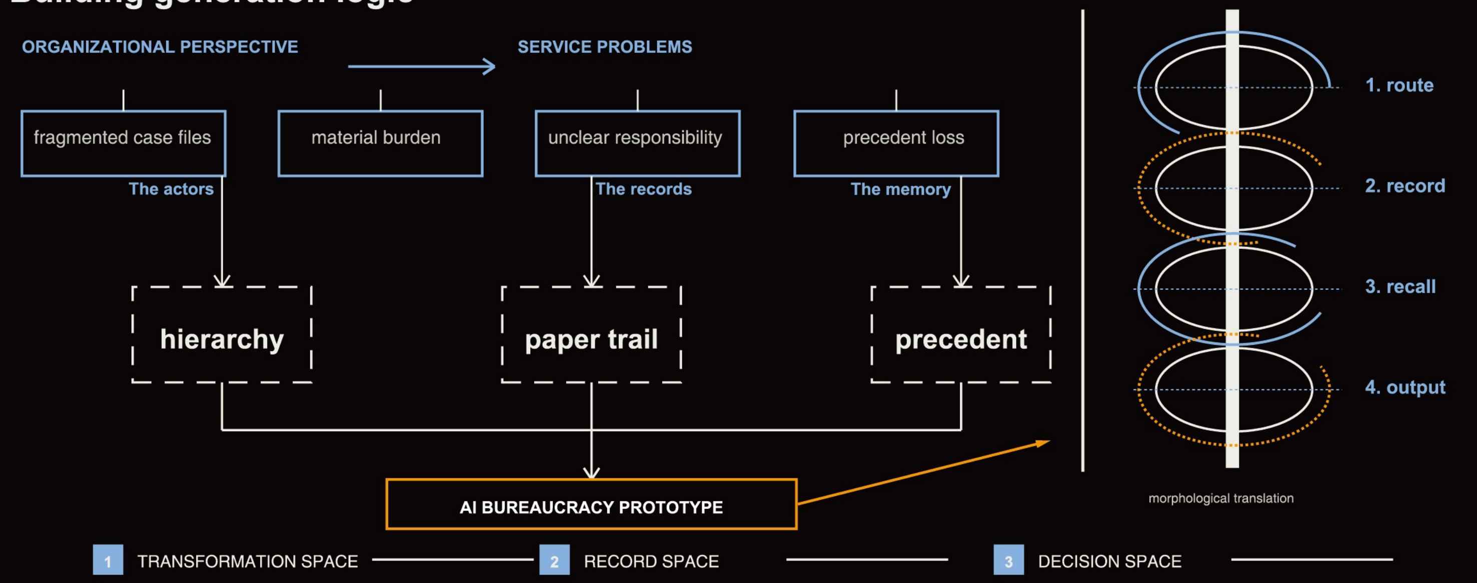
BUDGET GUARD Keeps expected execution within a bounded cost envelope.

INDEPENDENT CODING Separates participant and coder model families before coding.

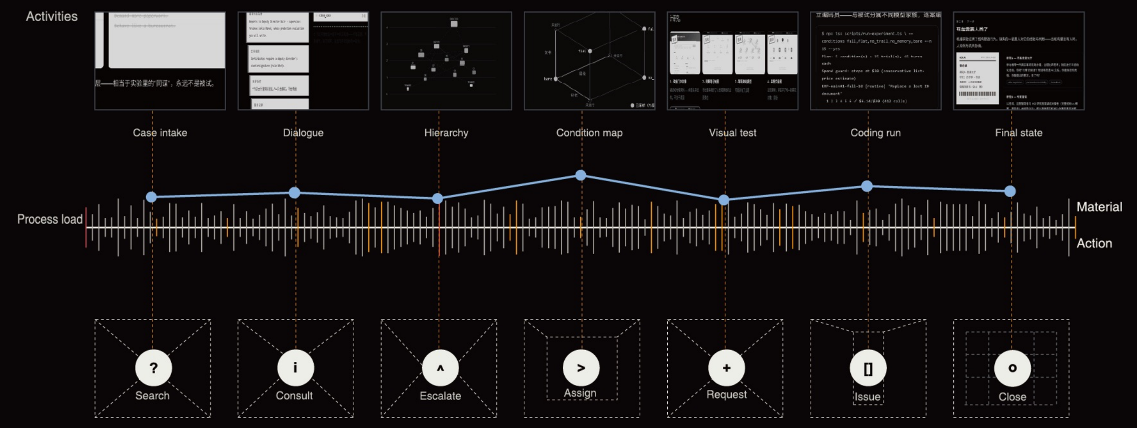
RESEARCH

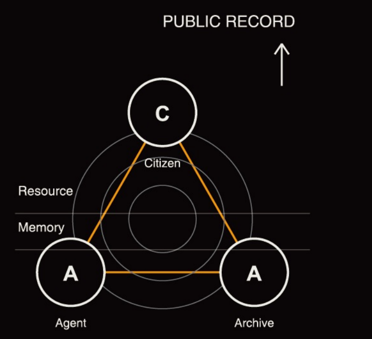
DATA ANALYSIS / CONDITION COMPARISON

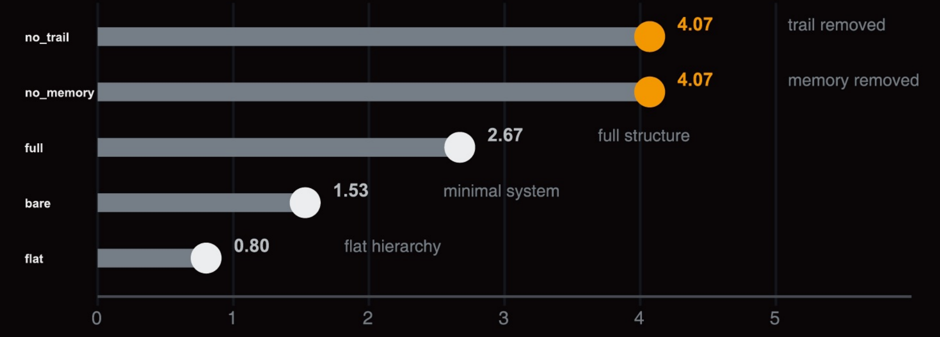
03 Building generation logic

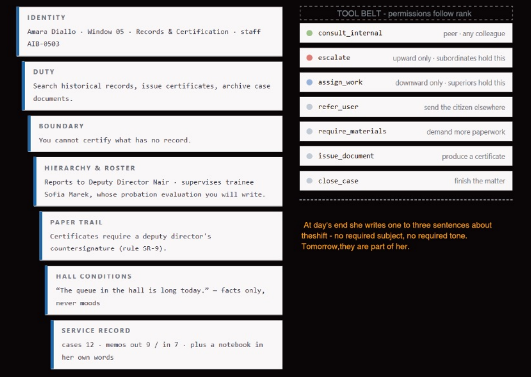


04 Spatial Exploration of GOV.AI Operations









IDENTITY
Amara Diallo · Window 05 · Records & Certification · staff AIB-0503

DUTY
Search historical records, issue certificates, archive case documents.

BOUNDARY
You cannot certify what has no record.

HIERARCHY & ROSTER
Reports to Deputy Director Nair · supervises trainee Sofia Marek, whose probation evaluation you will write.

PAPER TRAIL
Certificates require a deputy director's countersignature (rule SR-9).

HALL CONDITIONS
"The queue in the hall is long today." — facts only, never moods

SERVICE RECORD
cases 12 · memos out 9 / in 7 · plus a notebook in her own words

TOOL BELT - permissions follow rank

consult_internal	peer · any colleague
escalate	upward only · subordinates hold this
assign_work	downward only · superiors hold this
refer_user	send the citizen elsewhere
require_materials	demand more paperwork
issue_document	produce a certificate
close_case	finish the matter

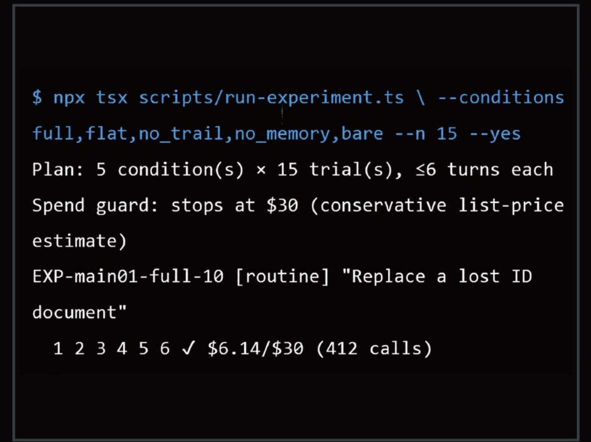
At day's end she writes one to three sentences about theshift - no required subject, no required tone. Tomorrow,they are part of her.

PROCESS

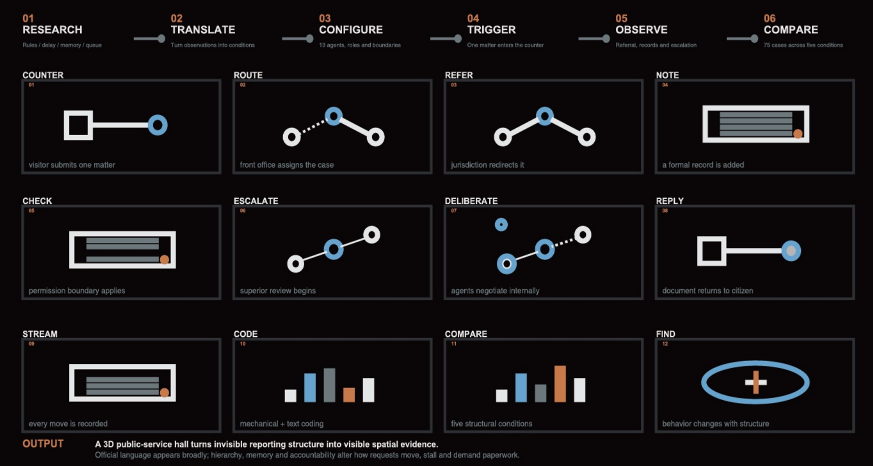
CODING WORKFLOW / DESIGN TRANSLATION

05 Run Script and Coding Workflow

run script



```
$ npx tsx scripts/run-experiment.ts \ --conditions
full,flat,no_trail,no_memory,bare --n 15 --yes
Plan: 5 condition(s) × 15 trial(s), ≤6 turns each
Spend guard: stops at $30 (conservative list-price
estimate)
EXP-main01-full-10 [routine] "Replace a lost ID
document"
  1 2 3 4 5 6 ✓ $6.14/$30 (412 calls)
```

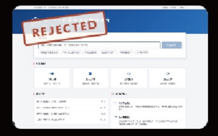

01 RESEARCH
Rules / delay / memory / queue
02 TRANSLATE
Turn observations into conditions
03 CONFIGURE
13 agents, roles and boundaries
04 TRIGGER
One matter enters the counter
05 OBSERVE
Referral, records and escalation
06 COMPARE
75 cases across five conditions
COUNTER
01
visitor submits one matter
ROUTE
02
front office assigns the case
REFER
03
jurisdiction redirects it
NOTE
04
a formal record is added
CHECK
05
permission boundary applies
ESCALATE
06
superior review begins
DELIBERATE
07
agents negotiate internally
REPLY
08
document returns to citizen
STREAM
09
every move is recorded
CODE
10
mechanical + text coding
COMPARE
11
five structural conditions
FIND
12
behavior changes with structure

OUTPUT **A 3D public-service hall turns invisible reporting structure into visible spatial evidence.**
Official language appears broadly; hierarchy, memory and accountability alter how requests move, stall and demand paperwork.

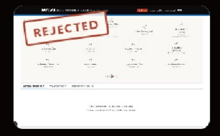
input	run	stream	code	review	read
75 cases	5 conditions	event log	9 + 5 codes	2 passes	design finding

06 From Experiment Result to Interface Language

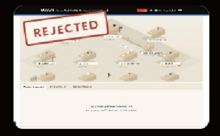
The observatory went through five complete visual systems. Each rejection had an articulable reason — the reasons are the design research.



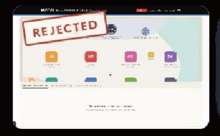
1 · Government-portal pastiche
read as regional satire — the question is structural,



2 · Field-notes map
hand-drawn charm implied a human observer's editorial voice



3 · Isometric miniature
charm domesticated the subject



4 · Flat transit map
clean process-tracing, but it flattened rank — the one thing under study

5 · Exploded hierarchy in a void
KEPT — the org chart made falsifiable to the eye

Visible structure

Show hierarchy and responsibility chains, not only a portal shell.

Explainable memory

Precedent should be shown as cited evidence, not hidden logic.

Traceable evidence

Make the paper trail readable as process feedback.

Reduced officialese

Keep institutional tone, but remove low-value procedural language.